

The strong thirteen spheres problem

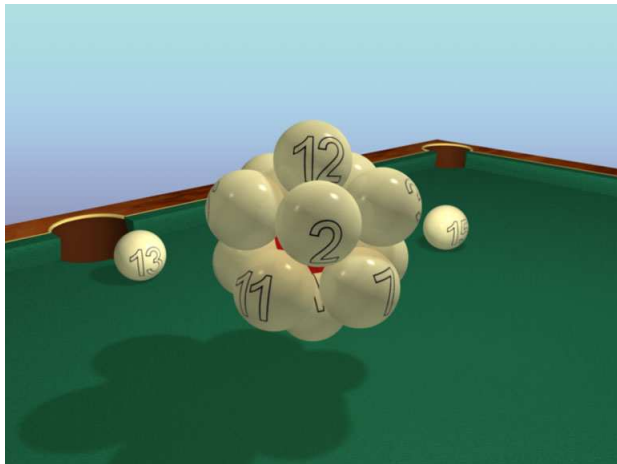
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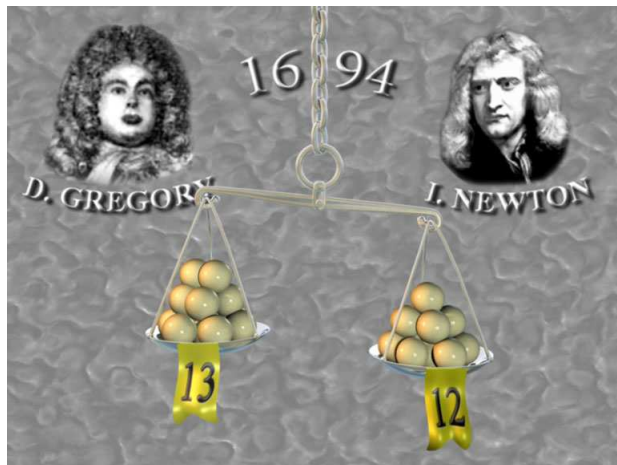
March 31, 2011

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The thirteen spheres problem



The thirteen spheres problem



The thirteen spheres problem: proofs

K. Schütte, and B. L. van der Waerden (1953)

John Leech (1956) : two-page sketch of a proof

... It also misses one of the old chapters, about the “problem of the thirteen spheres,” whose turned out to need details that we couldn’t complete in a way that would make it brief and elegant.

Proofs from THE BOOK, M. Aigner, G. Ziegler, 2nd edition.

W. -Y. Hsiang (2001);

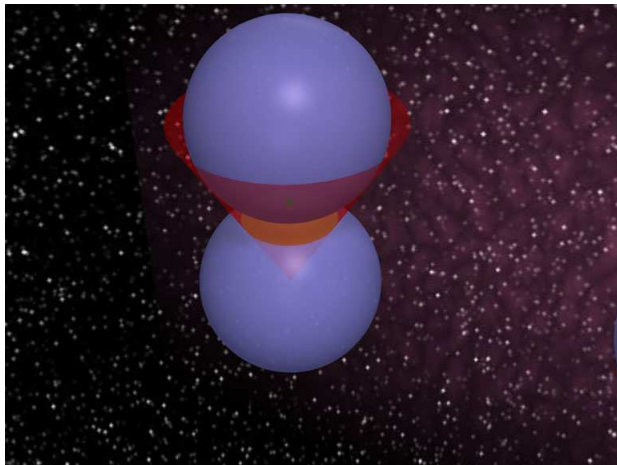
H. Maehara (2001, 2007);

K. Böröczky (2003);

K. Anstreicher (2004);

O. Musin (2006)

The Tammes problem



The Tammes problem

How must N congruent non-overlapping spherical caps be packed on the surface of a unit sphere so that the angular diameter of spherical caps will be as great as possible

Tammes PML (1930). "On the origin of number and arrangement of the places of exit on pollen grains". Diss. Groningen.

The Tammes problem

Let X be a finite subset of $\mathbb{S}^2$. Denote

$$\psi(X) := \min_{x,y \in X} \{\text{dist}(x,y)\}, \text{ where } x \neq y.$$

Then X is a spherical $\psi(X)$ -code.

Denote by d_N the largest angular separation $\psi(X)$ with $|X| = N$ that can be attained in $\mathbb{S}^2$, i.e.

$$d_N := \max_{X \subset \mathbb{S}^2} \{\psi(X)\}, \text{ where } |X| = N.$$

The Tammes problem

L. Fejes Tóth (1943) $N = 3, 4, 6, 12, \infty$

K. Schütte, and B. L. van der Waerden, (1951) $N = 5, 7, 8, 9$

L. Danzer (1963) $N = 10, 11$

R. M. Robinson (1961) $N = 24$

The Tammes problem for $N=13$

Lower bound:

$$d_{13} \geq 57.1367^\circ \text{ (see 3D model)}$$

Upper bounds:

$$d_{13} < 60^\circ \text{ Schütte and v. d. Waerden (1953)}$$

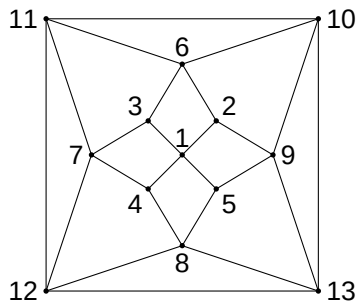
$$d_{13} < 58.87^\circ \text{ K. Böröczky and L. Szabó (2003)}$$

$$d_{13} < 58.50^\circ \text{ C. Bachoc and F. Vallentin (2008)}$$

Let X be a finite set in $\mathbb{S}^2$. The *contact graph* $\text{CG}(X)$ is the graph with vertices in X and edges (x, y) , $x, y \in X$ such that

$$\text{dist}(x, y) = \psi(X)$$

The contact graph $\Gamma_{13} := \text{CG}(P_{13})$ with $\psi(P_{13}) \approx 57.1367^\circ$



The value $d = \psi(P_{13})$ can be found analytically.

$$2 \tan \left(\frac{3\pi}{8} - \frac{a}{4} \right) = \frac{1 - 2 \cos a}{\cos^2 a}$$

$$d = \cos^{-1} \left(\frac{\cos a}{1 - \cos a} \right).$$

Thus, we have $a \approx 69.4051^\circ$ and $d \approx 57.1367^\circ$.

Theorem. The arrangement of 13 points P_{13} in $\mathbb{S}^2$ is the best possible, the maximal arrangement is unique up to isometry, and $d_{13} = \psi(P_{13})$.

Shift of a single vertex

Let X be a finite set in $\mathbb{S}^2$. Let $x \in X$ be a vertex of $\text{CG}(X)$ with $\deg(x) > 0$, i.e. there is $y \in X$ such that $\text{dist}(x, y) = \psi(X)$. We say that there exists a shift of x if x can be slightly shifted to x' such that $\text{dist}(x', X \setminus \{x\}) > \psi(X)$.

Danzer [1963] defined the following flip. Let x, y, z be vertices of $\text{CG}(X)$ with $\text{dist}(x, y) = \text{dist}(x, z) = \psi(X)$. We say that x is flipped over yz if x is replaced by its mirror image x' relative to the great circle yz . We say that this flip is *Danzer's flip* if $\text{dist}(x', X \setminus \{x, y, z\}) > \psi(X)$.

Irreducible graph

We say that the graph $\text{CG}(X)$ is *irreducible* [Schütte - van der Waerden, Fejes Tóth, and Danzer] (or *jammed*) if there are neither Danzer's flips nor shifts of vertices.

Let X be a subset of $\mathbb{S}^2$ with $|X| = 13$ and $\psi(X) = d_{13}$. Denote by G_{13} the graph $\text{CG}(X)$. Actually, this definition does not assume that G_{13} is unique. We use this designation for some $\text{CG}(X)$ with $\psi(X) = d_{13}$.

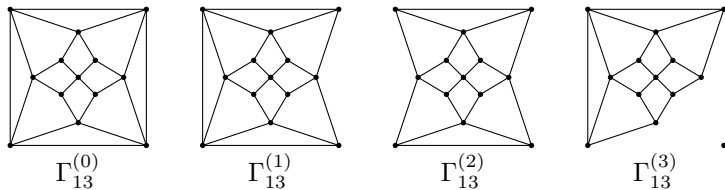


Figure: Graphs $\Gamma_{13}^{(i)}$.

Lemma 1. G_{13} is isomorphic to $\Gamma_{13}^{(i)}$ with $i = 0, 1, 2$, or 3 .

Lemma 2. G_{13} is isomorphic to $\Gamma_{13}^{(0)}$ and $d_{13} = \psi(P_{13}) \approx 57.1367^\circ$.

Properties of irreducible graphs

An irreducible graph for $N = 13$ satisfies the following properties:

- 1 The degree of a vertex = 0,3,4,5.
- 2 All faces are convex.
- 3 There are no faces with seven vertices and higher.
- 4 If there is an isolated vertex, then it lies in a hexagonal face.
- 5 No more than one vertex can lie in a hexagonal face.

The proof consists of two parts:

- (I) Create the list L_{13} of all graphs with 13 vertices that satisfy 1,3,4,5;
- (II) Using linear approximations and linear programming remove from the list L_{13} all graphs that do not satisfy the geometric properties of G_{13}

The list L_{13}

To create L_{13} we use the program *plantri* (Gunnar Brinkmann and Brendan McKay). This program is the isomorph-free generator of planar graphs, including triangulations, quadrangulations, and convex polytopes.

The program *plantri* generates 94,754,965 graphs in L_{13} . Namely, L_{13} contains 30,829,972 graphs with triangular and quadrilateral faces; 49,665,852 with at least one pentagonal face and with triangular and quadrilaterals; 13,489,261 with at least one hexagonal face which do not contain isolated vertices; 769,375 graphs with one isolated vertex, 505 with two isolated vertices, and no graphs with three or more isolated vertices.

Let G be a graph from the list L_{13} .

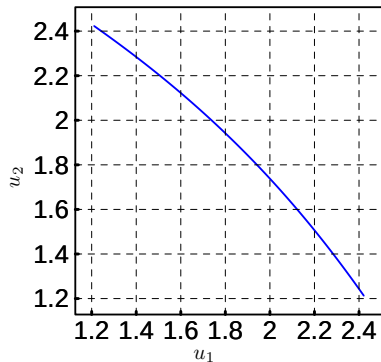
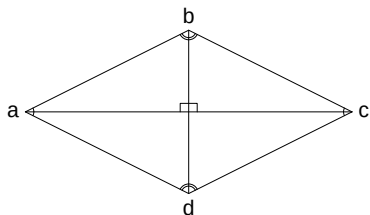
Variables: d (the length of edges), angles of faces.

Equations and inequalities:

- ❶ $d > 57.1367^0$.
- ❷ For each vertex sum of its angles $= 2\pi$.
- ❸ For a triangle: an explicit equation.
- ❹ For a quadrilateral: an explicit equation.
- ❺ For a pentagon: an approximation by linear inequalities.
- ❻ For an empty hexagon: an approximation by linear inequalities.
- ❼ For a hexagon with an isolated vertex: an approximation by linear inequalities.

$$u = \arccos \left(\frac{\cos d}{1 + \cos d} \right)$$

Quadrilateral



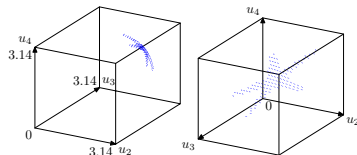


Figure: Space of admissible vectors $(u_1, u_2, u_3, u_4, u_5)$ of angles projected into (u_2, u_3, u_4)

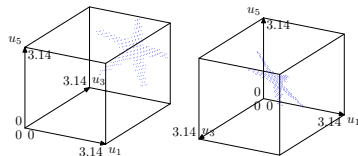


Figure: Space of admissible vectors $(u_1, u_2, u_3, u_4, u_5)$ of angles projected into (u_1, u_3, u_5)

Inequalities for a pentagon

$$u_{i-2} + u_{i+2} - 1.577u_i \geq 0 \quad (1)$$

$$\dots \quad (2)$$

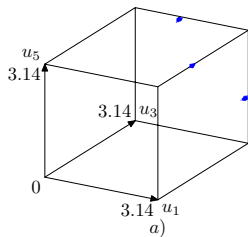
$$u_{i-2} + u_{i+2} - 0.9u_i \leq 2.71 \quad (3)$$

$$u_{i-1} + u_{i+1} + 1.6u_i \leq 8.56 \quad (4)$$

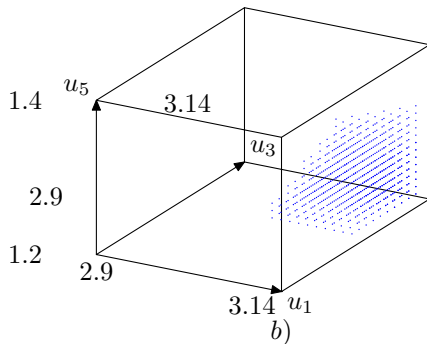
$$\dots \quad (5)$$

$$u_{i-1} + u_{i+1} + 1.55u_i \geq 7.83 \quad (6)$$

Empty hexagon



One component under the zoom



Linear inequalities (example)

Let u_i $0 \leq i \leq 5$ be angles of our hexagon. And let $u_0 < \pi/2$, then:

$$1.2 \leq u_0, u_3 \leq 1.34 \quad (7)$$

$$3.0 \leq u_1, u_2, u_4, u_5 \leq \pi \quad (8)$$

$$14.73 \leq \sum_{i=0}^5 u_i \leq 15.07 \quad (9)$$

Hexagon with an isolated vertex

$$\sum_{i=0}^5 u_i \geq 15.937 \quad (10)$$

Feasible solutions of the system

- 1 Do linear estimations of equations.
- 2 Using LP find a convex region containing a possible solution.
- 3 Using a region do more precise linear approximations and go back to steps 1,2.
- 4 If a region becomes empty – system is unfeasible.
- 5 if a region is still not empty split it into two parts, and go back to steps 1–5.
- 6 If all regions (after splitting) become empty – system if unfeasible.

Three feasible graphs

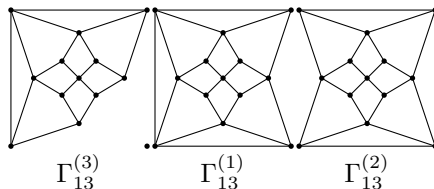


Figure: Three graphs which are survived: with a hexagon containing an isolated vertex, with a pentagon, with an empty hexagon

Proof of Lemma 2

Take some graph write all equations for angles u_i . Number of equations unit less than number of variables. Fix d and solve set of equations. Obtained solution satisfy all inequalities only then $d \leq d_{13}$.

